Human Computer Interaction
TICT 2153 (T)

Ergonomic Posture Monitor for Computer Users

Group No: Group 05

Date of Submission: 2026/06//25

MINI GROUP PROJECT - PROPOSAL SUBMISSION

1. Group Members

	Registration Number	Name	Role in Project
1	2023/ICTS/38	I .D.T.Chathubhashana	Hardware Design
2	2023/ICTS/39	H.D.H. Erandaka (L)	Team Leader
3	2023/ICTS/40	T.M.D Madusanka	Circuit Design
4	2023/ICTS/43	G G S Chanuka	Testing
5	2023/ICTS/44	A.T Dilhari	Presentation
6	2023/ICTS/45	F.M.F.Faara	Documentation
7	2023/ICTS/46	W.A.D.M. Weerasinghe	Prototype Development

Note: Mention the team leader

2. Problem Statement

Prolonged computer use and poor sitting posture are leading causes of musculoskeletal complications, chronic back pain, and spinal strain among university students and desk-bound office workers. While multiple desktop software programs and mobile application reminders exist to encourage better posture, they suffer from a major limitation: they are easily dismissed, muted, or hidden beneath active windows, causing digital "notification fatigue." There is a critical lack of an ambient, physical, and tangible workspace companion that continuously reminds users of their physical form. A non-intrusive, physical intervention is needed to seamlessly build healthy long-term habits without causing cognitive disruption during intense computer tasks.

3. Target Users

- **Primary Users:** University IT/Engineering students, software developers, and administrative corporate workers aged 18 to 45 who stay seated for more than 4 hours a day in front of a computer screen.

- **Abilities & Needs:** Cognitively and physically capable individuals who require external passive physical prompts to correct subconscious slouching tendencies.

- **Context of Use:** Stationary indoor desktop workspaces, such as university computer laboratories, quiet library study cubicles, corporate offices, and home study desks.

4. Proposed Solution & Key Features

The **Ergonomic Posture Monitor** is a physical desktop companion powered by an **ESP32-CAM module**. Placed on the workstation, it utilizes its integrated camera to monitor the user's upper-body alignment and distance in real-time. To prevent user annoyance and notification fatigue, the device implements a **3-tier progressive feedback mechanism** based on the duration of poor posture:

- **Tier 1 (Good Posture):** As long as the user maintains an ergonomic upright posture, the RGB LED glows **Green**, indicating a safe state.
- **Tier 2 (Temporary Bad Posture / Soft Warning):** If the user slouches or leans forward, the RGB LED immediately transitions to **Yellow/Blue** for a brief buffer period (**up to 3 minutes**). This acts as a silent, ambient visual prompt, giving the user a chance to self-correct during brief desk movements without an annoying sound.
- **Tier 3 (Persistent Bad Posture / Critical Escalation):** If the user continuously maintains the bad posture for **over 3 minutes**, the device escalates the alert. The RGB LED turns **Red**, and the **Active Buzzer** emits a brief audible nudge to ensure compliance.

1. **Real-Time Visual Posture Tracking:** Continuously monitors facial and shoulder alignment using the ESP32-CAM.
2. **3-Tier Progressive LED Feedback:** Shifts ambient colors dynamically (Green, Yellow, Red) based on slouch duration.
3. **Time-Delayed Auditory Alert:** Triggers the active buzzer only after a persistent 3-minute slouch threshold.
4. **Standalone Hardware Independence:** Operates entirely as a physical desk object without needing computer software.

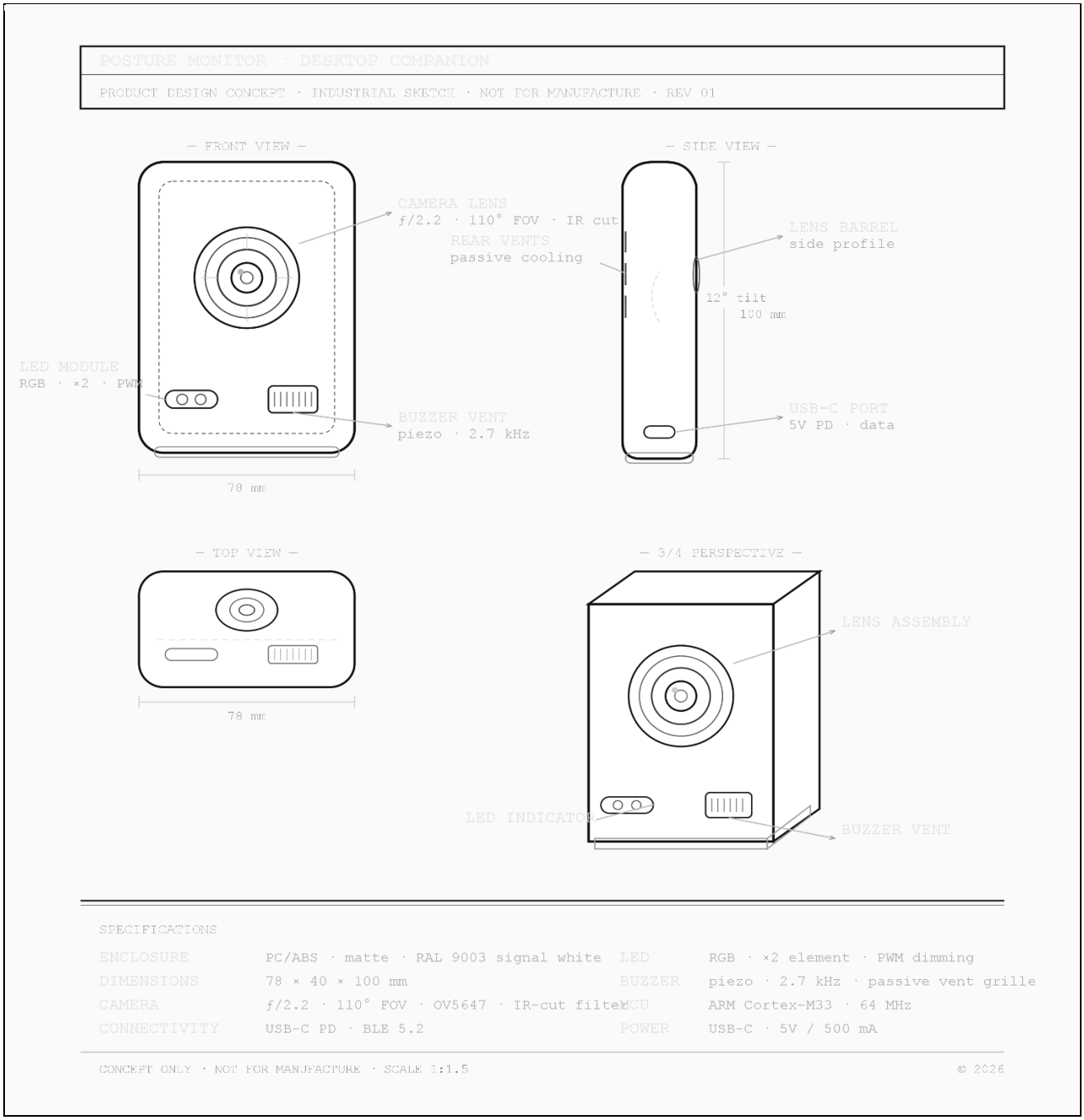
5. Design Thinking Plan

Stage	Planned Activities	Responsible Member(s)
1 - Empathize	Interview students and computer users about posture-related issues and physical discomfort.	All Members
2 - Define	Identify and analyze the main causes and long-term effects of poor desk posture.	Team Leader & Documentation Team
3 - Ideate	Brainstorm possible hardware solutions and select the best progressive alert design layout.	All Members
4 - Prototype	Build the ESP32-CAM hardware circuit, program the core logic, and assemble all components.	Hardware & Circuit Team
5 - Test	Test the tracking accuracy, user comfort during the 3-minute buffer, and final alert effectiveness.	Testing Team

6. Feasibility & Expected Impact

Materials / Components / Tools Required	Expected Impact / Who Will Benefit
Hardware Components: • ESP32-CAM Board & OV2640 Camera • ESP32-CAM MB Programmer Shield • Micro-USB Data Cable (for code upload & power) • RGB LED Module • Active Buzzer Module • Breadboard & Jumper Wires Software & Enclosure • Arduino IDE & ESP32 Camera Libraries • Compact Plastic or Acrylic Casing	Target Beneficiaries: • University students, software developers, and desk-bound office workers sitting for over 4 hours. Health & Ergonomic Impact: • Promotes active upper-body posture self-correction. • Reduces long-term risks of neck pain, backache, and spinal fatigue. HCI & Behavioral Value: • Smart 3-minute alert delay prevents user annoyance. • Eliminates digital "notification fatigue" via physical desk alerts.

7. Product Sketch (Hand-drawn or Digital)



8. Submission Checklist (Tick before submitting)

□	Project title is clear and descriptive
□	All group member names, Registration numbers, and roles are filled in
□	Problem statement is specific and references real users
□	Proposed solution describes a physical/tangible product (NOT software/app)
□	At least 3 key features are listed
□	Design Thinking plan is completed with stage-by-stage responsibilities
□	Materials / components required are listed under Feasibility
□	A sketch of the product is included in Section 7

Lecturer's Signature: _____ *Date:* _____